Project Outline

Project context

Basically track fabric samples in & out. All data is on prep sheet. The prep sheet has info abt this.

- Here is a paste from my communications with seward:
  You showed us 3 conditioning rooms, could you remind us of the temperature, humidity, and tolerances of each room? Also, it'd be great to get in writing an example of how each room is used and if they used the same labeling workflow.
- The dedicated (smaller) room for PPE is 23C +/-3 C and 65% RH +/-5%RH
  - This room is dedicated for sample(fabric, gloves, footwear, and latex gloves) conditioning. Samples are brought in and placed on racks for a minimum of 24 hrs prior to removal for testing.
- The shared room with the larger door is 23C +/-1C and 65% RH +/-2%RH
  - This room is dedicated for sample(fabric, gloves, footwear, Ropes, garments and latex gloves) conditioning. Samples are brought in and placed on racks for a minimum of 24 hrs prior to removal for testing.
- The interior room is 21C +/-3C and 50%RH +/- 5%RH
  - This room is dedicated for footwear sample conditioning. Samples are brought in and placed on racks for a minimum of 24 hrs prior to testing.
- Can you clarify the terminology of "sample" vs "specimen"?
  - A sample is a set or subset of specimens for a particular model
  - A specimen is one individual test piece
- The top right of a label contains the pre-conditioning of a sample, whether that is "as received" (AR) or 5L (5x laundered) 50L (50x laundered). What are the other options? Can you specifically say how many laundering options there are and any additional codes?
  - There is "AR" for as received, and we have these laundry cycle options: 1L, 3L, 5L, 10L, 25L, 50L and 100L.

Typically samples are uniquely identified by these characteristics: a 12 digit project number (which is associated with a single job, which can have multiple fabrics and tests), the specific test that is done, and whether that fabric sample is laundered or unlaundered (as above, we can probably fit this in two digits or just a number could be negative etc or 0 for AR, but there are options in addition to launering, like it could be wetted or like heated). Bc for each sample (it's a stack) they will cut the fabrics up into specimens for the thing so it can be repeated.

The issue is that sometimes people will forget to log in (bc the room is a separate room and they have to go to the word doc for each thing and update the thing) and now they have no evidence that it was in there for >24hr even tho it was and they don't want to fudge the log so it wastes them a lot of time.

System architecture

So what we want to do is first. They already have a label printer that includes the info: project nr, test, laundering, sample/specimen. So we would want to put this in a machine readable format. It could be a UUID. It could be encoded base64. We haven't decided – help us decide. It's weird database shit like does this printing subsystem necessarily need that? We don't know. But essentially this subsystem is it takes the pdf (or document) of the prep sheet, and turns this into labels. They already have label printing but that's only text. We are okay with making a new label printer. Currently, they have an old Dymo label printer. But what format of barcode? Rfid? We don't know. Best way to store this data? I think we should have a web app. It should take the pdf in, and then generate a csv intermediate of all the labels that should be printed. But is it even possible to create this architecture? Should we paste the csv into a ready built app that is supplied by the label printer manufacturer, or should it can it be an all in one exe file that takes pdf and labels come out?

The next problem is split between two people. Database backend & database frontend. Sometimes they won't take out the entire stack at once. We don't know if we want barcodes on the entire thing. Right now they label every single item in a stack, and manually write numbers outside of the printed areas (project number; laundering status) so they have to note sample / test / something else. We don't necessarily need barcodes and rfid on every single item, it could just be on the top. so default behavior should be like when you scan a stack in or out if you scan anything from the stack (or the top if its only one, but if its any from a stack it should assume it's all coming out @ the same time) it should scan all specimens in that stack. But for the frontend, it should address two areas. The first is the machine at the point of scan. Like it should have a screen (touchscreen) to display (in/out choice, but it doesn't necessarily have to do that it could just assume that 2nd scan was out, then it should also have a choice relative to maybe for edge cases whether scan individually or a whole stack, and perhaps an undo button. It should give a red button screen when its offline to show the transactions won't complete. Then also it will have a green when the scan succeeds (like when the DB transaction goes to remote and is confirmed). Wait then also should the DB record raw transaction logs ONLY? Or just the db should include one object for each specimen and in & out time?

Lastly is like the physical shit.

Subsystems

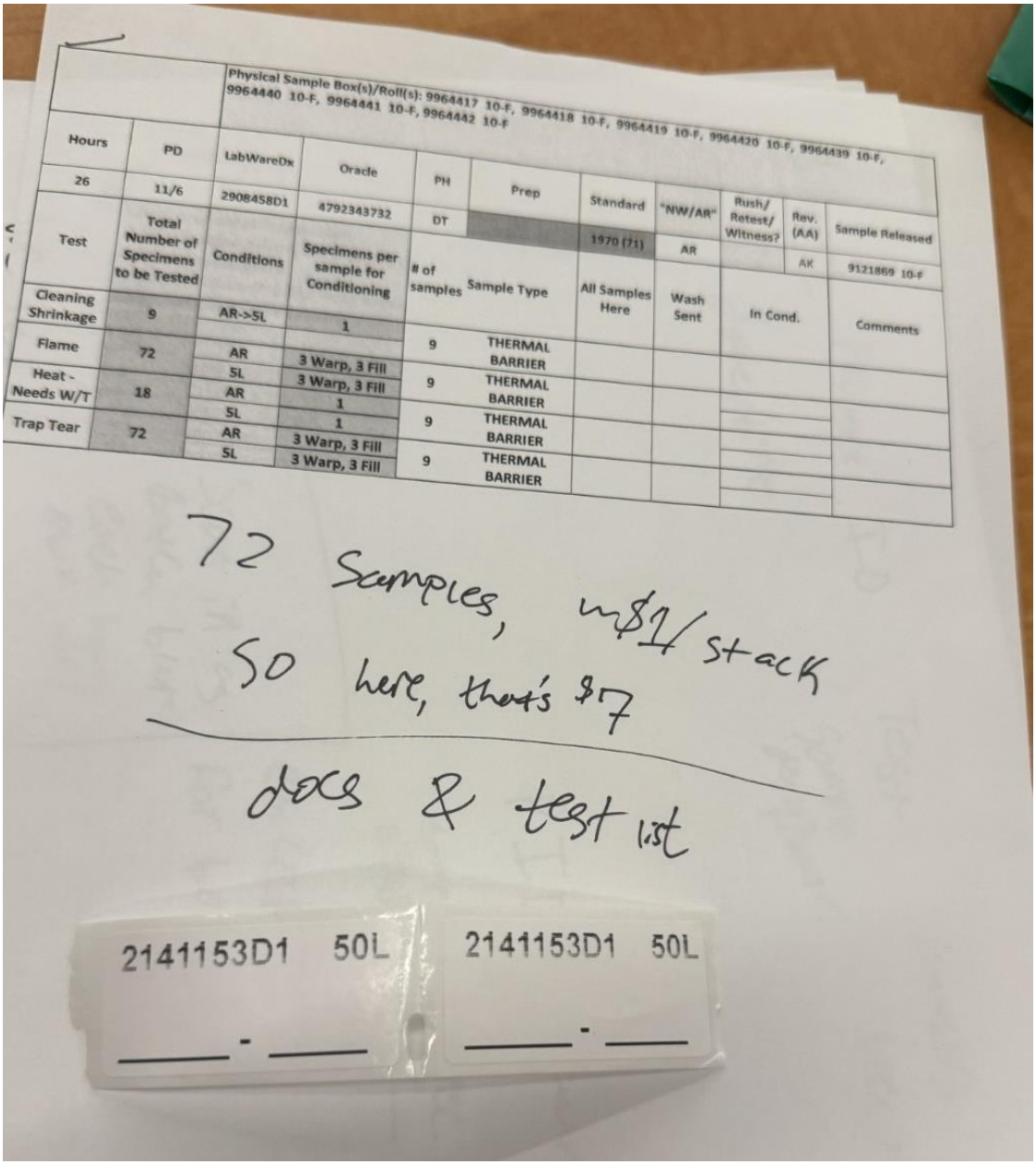
Lwk they’re kind of defined above

Team roles

1 person does physical, 1 does db front, 1 does db back, 1 does the label print

Clarifications

So there’s some departures from the technical memos below. I don’t really give a fuck about network outages now. I think that it should just say NOT AVAILABLE.



Physical Sample Box(s)/Roll(s): 9964417 10-F, 9964418 10-F, 9964419 10-F, 9964420 10-F, 9964439 10-F, 9964440 10-F, 9964441 10-F, 9964442 10-F

Hours	PD	LabWareDx	Oracle	PH	Prep	Standard	"NW/AR"	Rush/ Retest/ Witness?	Rev. (AA)	Sample Released
26	11/6	2908458D1	4792343732	DT		1970 (71)	AR		AK	9121869 10-F

Test	Total Number of Specimens to be Tested	Conditions	Specimens per sample for Conditioning	# of samples	Sample Type	All Samples Here	Wash Sent	In Cond.	Comments
Cleaning Shrinkage	9	AR->5L	1						
Flame	72	AR	3 Warp, 3 Fill	9	THERMAL BARRIER				
		5L	3 Warp, 3 Fill						
Heat - Needs W/T	18	AR	1	9	THERMAL BARRIER				
		5L	1						
Trap Tear	72	AR	3 Warp, 3 Fill	9	THERMAL BARRIER				
		5L	3 Warp, 3 Fill						

72 Samples, m$1/stack

50 here, that’s $7

docs & test list

2141153D1 50L _____-_____

2141153D1 50L _____-_____

^ this image is an example of a prep sheet on the top, and on the bttom is two labels with the top left being project number and top right being the laundering status. Then we can create letter codes for the specific test used.

I think a pi 5 with touchscreen is not a bad slution. But it should really be plug and play. If the power goes out and it restarts it should go straight into that app and not have anything else.

I have pasted two technical memos below – but some is just bullshit essentially just to meet the assignment details.

To Michael Seward
From Team FFF: William Zhang, Max Rerisi, Raquibul Hasan, Dereknick Garcia-Morales
8 September 2026
Problem Statement and Design Criteria

Problem Statement

UL Solutions laboratory staff need an automated and auditable method to track groups of test samples as they enter and leave conditioning rooms. The current process requires employees to manually record sample movement and timestamps, creating opportunities for missed or incorrect entries, and unnecessary work.

Importance and Need

Before testing, material samples must remain in controlled temperature and humidity conditions for the required conditioning period (at least 24 hours). Reliable records of when samples enter and leave the conditioning room are therefore necessary to demonstrate that the conditioning process was followed and to support subsequent audits.

Currently, employees enter this information into each individual project's datasheet. This means when putting large amounts of samples into the conditioning room, the employee must enter and exit the room to enter the data into their laptop. This also means that the employee is using multiple Word documents for each datasheet simultaneously, creating mental overhead. Missed or incorrect conditioning timestamps can compromise the audit trail and may require UL to investigate or repeat testing.

An automated system would need to check items in and out to reduce manual data entry while keeping an auditable log of sample tracking times. It's also important that the system accommodates edge cases; employees should be able to adjust the records easily.

Users and Clients

The client is UL Solutions (Travis Westphal/Michael Seward). The primary users are UL laboratory employees who prepare, condition, and test material samples. Audit personnel are secondary users of the system because they audit the logs it creates.

Function

The proposed system will track a stack of related test material as a single unit during normal conditioning operations. Items within a stack share the same project number, conditioning status, and test type and normally enter and leave conditioning together. A machine-readable label containing these three fields will be placed on the top of the stack.

When the label is scanned, the system will identify the project, conditioning status, and test; determine whether the stack is entering or leaving the conditioning room; automatically record the date and time; and provide visible confirmation that the transaction was successfully recorded. Routine operation will require one scan and no manual data entry.

The system will maintain two related forms of data. The first will be an automated record of the transactions detected by the scanner. The second will provide the detailed records required for UL's existing audit process. Authorized users will be able to modify the audit-facing record to address exceptional cases while preserving traceability of the original transaction and subsequent changes.

Design Criteria

Table 1 summarizes the objectives and constraints used to evaluate the sample-tracking system.

Table 1: Criteria for Sample Time Tracking System

Objective, Constraint	Target value / performance criterion	Justification
Auditable database (objective)	Record project number, conditioning status, test type, IN/OUT status, and timestamp for each transaction. Maintain auditable history, including corrections, for ≥5 years.	Auditability is the client's highest priority. Exceptional cases require records to be corrected without losing the history of what was originally recorded.
Scan accurately (objective)	Correctly read and record ≥99% of labels on the first scan under normal operating conditions.	The current process is vulnerable to incorrect or missed entries; the system is only useful if it's correct.
Web-outage resilience (objective)	Continue accepting and locally storing transactions for ≥24 hours without internet access, synchronizing after reconnects.	A temporary network outage should not force employees to return to manual recording or lose records of samples.
Track multiple fabric items at	A single scan must register one stack containing up to 20 test items without individually scanning each item.	Samples within a conditioning stack normally share the same project, conditioning status, and test and move

once (objective)		through conditioning together, requiring only a single scan.
Scan quickly / ease of use (objective)	Provide visible success confirmation within 1 s of the scan trigger. Normal operation should require one scan and zero typed entries per stack.	The system should not disrupt the existing laboratory workflow. Immediate confirmation also allows an employee to identify a failed scan before leaving the scanner.
Cost (constraint)	Total initial system cost <$5,000 and recurring consumables ≤$1 per stack.	UL gives us these numbers. Reliability and auditability are worth a certain per-unit cost to them as the alternative is redoing tests.
Operating environment (constraint)	All systems and components are fully functional in a room at 23 ± 1 °C and 65 ± 5% RH.	The equipment must function reliably in the conditioning room.

Our Pairwise Comparison Chart (Appendix I) ranked the objectives from highest to lowest as follows: auditable database, scan accurately, web-outage resilience, track multiple fabric items at once, and scan quickly / ease of use.

Appendix I: Pairwise Comparison Chart

	Scan quickly / ease of use	Scan accurately	Track multiple fabric items at once	Auditable database	Web-outage resilience	Totals
Scan quickly / ease of use	--------	0	0	0	0	0
Scan accurately	1	-------	1	0	1	3
Track multiple fabric items at once	1	0	-------	0	0	1
Auditable database	1	1	1	-----	1	4
Web-outage resilience	1	0	1	0	-------	2

To Michael Seward
From Team FFF: William Zhang, Max Rerisi, Raquibul Hasan, Dereknick Garcia-Morales
28 September 2026

Evaluation of Design Decisions

Overview

At UL Solutions, laboratory technicians evaluate fabric and material samples that require mandatory environmental conditioning for a minimum of 24 hours at 23 ± 1 °C and 65 ± 5% relative humidity prior to standardized testing. Currently, technicians record the entry and exit timestamps of sample stacks manually across separate project datasheets in Microsoft Word. This manual process introduces significant mental overhead, risks missed or inaccurate entries and threatens the integrity of UL's audit trail. UL Solutions requires an automated, resilient, and auditable tracking system that logs groups of up to 20 samples with a single scan while eliminating manual data entry.

To determine the optimal system design, our team decomposed the tracking problem into three functional subsystems: Identification Medium, Scanning Architecture, and Database Architecture. We brainstormed 17 total subsystem concepts and filtered them using Pugh screening matrices against baseline reference standards. Surviving concepts were mapped into a morphological chart, producing five fully integrated design solutions (Solutions A through E). We then evaluated these candidate solutions using a weighted Pugh scoring matrix based on design criteria established with the client.

Our evaluation identified Solution E (Fixed RFID Reader Station) as the winning design with an overall score of 4.65 out of 5.00. Solution E pairs a fixed desktop/shelf radio-frequency identification (RFID) reader at the conditioning room entryway with passive ultra-high-frequency (UHF) RFID tags affixed to each sample stack, maximizing first-scan accuracy and operational speed while minimizing equipment wear.

Ideation/Selection Process

To systematically explore potential solutions, our team decomposed the automated tracking system into three primary functional blocks: Identification Medium (6 brainstormed concepts), Scanning Architecture (6 brainstormed concepts), and Database Architecture (5 brainstormed concepts). Concepts ranged from optical character recognition (OCR) and Bluetooth Low Energy (BLE) beacons to overhead conveyor gantries and cloud-native databases.

Each concept was evaluated within its respective functional block using separate Pugh screening matrices (see Appendix, Table 2.1, Table 2.2, and Table 2.3). Concepts were evaluated against conventional laboratory baselines—specifically 1D printed barcodes, handheld optical scanning guns, and local spreadsheet logging—using binary scoring (+, 0, −) across client criteria. From this screening stage:

- Identification Medium was narrowed to two viable technologies: Barcodes (retained for low consumable cost) and Passive UHF RFID (selected for non-line-of-sight capability).

- Scanning Architecture was narrowed to three configurations: Handheld readers, Fixed desktop/shelf scanners, and Walk-through portal scanners.
- Database Architecture converged on a unified architecture: a relational SQL database with local SQLite caching. Because all viable solutions must satisfy the 5-year audit logging and 24-hour network resilience constraints established in Technical Memo #1, database design was standardized across all hardware options and removed as an independent variable from the final concept combinations.

We synthesized the surviving hardware blocks using a Morphological Chart (Appendix, Table 2.4), yielding five distinct, fully integrated candidate designs (Solutions A through E). These five solutions advanced to the weighted Pugh scoring matrix.

Final Design Solutions

- **Solution A (Barcode Handheld):** Utilizes a corded or wireless handheld optical scanner to manually target and read a printed 1D/2D barcode sheet positioned on top of the fabric stack.
- **Solution B (Barcode Fixed):** Employs a fixed presentation barcode scanner mounted at waist height, requiring technicians to orient and slide the top barcode label beneath the scanning window.
- **Solution C (RFID Handheld):** Uses a mobile, pistol-grip UHF RFID reader to scan an RFID inlay tag attached to the sample stack from a short range without line-of-sight constraints.
- **Solution D (RFID Walk-Through):** Integrates doorway RFID portal antennas and motion sensors that automatically detect tagged sample stacks as personnel walk across the room boundary.
- **Solution E (RFID Fixed):** Features a stationary desktop/shelf RFID scanning pad positioned at the conditioning room entrance where sample stacks are placed or passed over during check-in/check-out.

Pugh Scoring Matrix

Table 1.1 displays the weighted Pugh scoring matrix used to evaluate Solutions A through E. Criteria weights were determined using our team's initial Pairwise Comparison Chart, refined to reflect the primary operational differentiators between hardware configurations.

Table 1.1: Weighted Pugh Scoring Matrix

Design Criteria	**Weight**	**A**		**B**		**C**		**D**		**E**	
		Rating	Score	Rating	Score	Rating	Score	Rating	Score	Rating	Score
First-Scan	40%	3	1.20	2	0.80	5	2.00	2	0.80	5	2.00

Accuracy											
Speed	30%	3	0.90	2	0.60	4	1.20	5	1.50	5	1.50
Durability	25%	2	0.50	4	1.00	3	0.75	5	1.25	4	1.00
Cost	5%	5	0.25	5	0.25	3	0.15	1	0.05	3	0.15
Total Score	**100%**	—	**2.85**	—	**2.65**	—	**4.10**	—	**3.60**	—	**4.65**
Rank	—	—	**4**	—	**5**	—	**2**	—	**3**	—	**1**
Continue	—	—	**Yes**	—	**No**	—	**No**	—	**No**	—	**Yes**

Selected Solution

Upon completing the Pugh screening, morphological synthesis, and weighted scoring evaluations, our team decided to advance both Solution E (Fixed RFID Reader Station) and Solution A (Handheld Barcode Scanner) in parallel into physical prototyping. In our weighted Pugh matrix, Solution E earned the highest overall score of 4.65 out of 5.00, driven by top marks of 5 in our two most heavily weighted criteria: first-scan accuracy (40% weight) and speed (30% weight) (see Table 1.1). By utilizing passive UHF RFID near-field interrogation, Solution E eliminates optical line-of-sight requirements, allowing a technician to register an entire 20-sample stack in under half a second without adjusting orientation. Furthermore, integrating the scanner into a stationary entryway surface pad protects the sensitive hardware from repetitive drops, impacts, and cable fatigue, earning a durability score of 4.

Despite Solution E's technical advantages, cost remains a significant practical constraint for laboratory deployment. Dedicated UHF RFID interrogators, near-field antennas, and continuous tag purchases carry a much higher capital and consumable expenditure, reflected in Solution E's rating of 3 for cost. In contrast, Solution A earned the maximum score of 5 in cost. Off-the-shelf optical barcode imagers are inexpensive, and printing traveler sheets on standard paper stock incurs virtually zero recurring expense ($0.01 per stack). While Solution A scored lower in speed (3) and first-scan accuracy (3) because technicians must actively aim the scanner at the top label, it still satisfies all baseline project requirements.

Advancing both designs simultaneously offers the best balance between system performance and project risk. More than 90% of the underlying system architecture is completely shared between Solution A and Solution E: both utilize the exact same local SQLite database schema, 24-hour offline queuing engine, tamper-resistant audit logging rules, and automated server synchronization scripts. The only technical difference is the physical data-capture input—reading incoming keystrokes from an optical scanner versus parsing serial packets from an RFID reader. Because Solution A is far easier to assemble and debug quickly, our team will build it out first as an operational baseline to immediately eliminate the manual Word datasheet workflow.

Developing Solution E alongside it allows us to test both physical interfaces side by side in the conditioning room, giving UL Solutions the exact empirical data needed to determine whether the speed and convenience of RFID warrant the higher hardware investment.

Appendix

Table 2.1: Identification Pugh Screening Matrix

Design Criteria	OCR	1D Barcode	2D Barcode	NFC Tag	Passive RFID	BLE Beacon
First-Scan Accuracy	−	0	+	+	+	0
Scan Speed	−	0	0	0	+	+
Stack Tracking	−	0	0	0	+	+
Consumable Cost	+	0	0	−	−	−
Environmental Resilience	−	0	0	+	+	0
Total Score	**−3**	**0**	**+1**	**0**	**+2**	**+1**
Advance Concept?	**No**	**Yes**	**Yes**	**No**	**Yes**	**No**

Table 2.2: Scanning Pugh Screening Matrix

Design Criteria	Handheld Scanner	Fixed Scanner	Walk-through Portal	Automated Conveyor	Smartphone Camera App
Ease of Use / Workflow Speed	0	+	+	+	−
Hardware Durability	0	+	+	0	−
Installation Feasibility	0	0	−	−	+
Capital Equipment Cost	0	0	−	−	+
False-Trigger Avoidance	0	0	−	0	−
Total Score	**0**	**+2**	**−1**	**−1**	**−1**

Advance Concept?	**Yes**	**Yes**	**Yes**	**No**	**No**

Table 2.3: Database Pugh Screening Matrix

Design Criteria	Excel / SharePoint	Enterprise Platform	Central Cloud SQL	Local SQLite + Cloud Sync
5-Year Audit Trail Integrity	0	+	+	+
Offline Resilience (≥24 hr)	0	−	−	+
Implementation Complexity	0	−	0	0
System Licensing Cost	0	−	0	+
Total Score	**0**	**−2**	**0**	**+3**
Advance Concept?	**No**	**No**	**No**	**Universal**

Table 2.4: Morphological Chart & Solution Selections

Design Block	Option 1	Option 2	Option 3
Scanning Code	Plain Text (Status quo / OCR)	Barcode (1D / 2D)	RFID Tag (Passive UHF)
Scanning Hardware	Handheld Scanner (moves to pile)	Fixed / Stationary Scanner (pile brought to scanner)	Walk-Through Scanner (doorway portal)
Database System	Excel / SharePoint (Current datasheets)	Custom SQL Solution (Local SQLite + Central DB)	Enterprise DBMS (Palantir / Databricks)

- Solution A: 2 (Barcode); 1 (Handheld); 2 (SQL)
- Solution B: 2 (Barcode); 2 (Fixed); 2 (SQL)
- Solution C: 3 (RFID); 1 (Handheld); 2 (SQL)
- Solution D: 3 (RFID); 3 (Walk-through); 2 (SQL)
- Solution E: 3 (RFID); 2 (Fixed); 2 (SQL)